

SOLID PLASTIC AND E-WASTE RULES

THREE-STREAM FIREWALL → RULE-VINTAGE LADDER → MUNICIPAL RESPONSIBILITY MAP → SEGREGATION-TO-DISPOSAL CHAIN → PLASTIC ACTOR MATRIX → BAN-VERSUS-EPR GATE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00

THREE-STREAM FIREWALL

THREE-STREAM FIREWALL

SOLID WASTE

MUNICIPAL GENERATION, SEGREGATION AND LOCAL SERVICES

PLASTIC WASTE

PRODUCT AND PACKAGING CONTROLS PLUS EPR

ELECTRICAL AND ELECTRONIC END-OF-LIFE CHAIN

COMMON PARENT

ENVIRONMENT PROTECTION ACT FRAMEWORK

ECOLOGICAL UNIT / PARAMETER

SOLID WASTE → municipal generation, segregation and local services

PLASTIC WASTE → product and packaging controls plus EPR

DECISIVE DISTINCTION

E-WASTE → electrical and electronic end-of-life chain

COMMON PARENT → Environment Protection Act framework

SCALE / STATUS / EXCEPTION

RULE → separate streams remain legally distinct

SOLID WASTE → municipal generation, segregation and local services

ECOLOGICAL UNIT / PARAMETER

• SOLID WASTE → municipal generation, segregation and local services

• PLASTIC WASTE → product and packaging controls plus EPR

DECISIVE DISTINCTION

• E-WASTE → electrical and electronic end-of-life chain

• COMMON PARENT → Environment Protection Act framework

SCALE / STATUS / EXCEPTION

• RULE → separate streams remain legally distinct

• SOLID WASTE → municipal generation, segregation and local services

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PLASTIC WASTE → product and packaging controls plus EPR.

01

STAGE 01

RULE-VINTAGE LADDER

RULE-VINTAGE LADDER

BASE RULE

IDENTIFY NOTIFICATION

IDENTIFY CHANGED PROVISION AND DATE

GUIDELINE OR SOP

IMPLEMENTATION INSTRUCTION

PORTAL PROCEDURE

DIGITAL COMPLIANCE WORKFLOW

BASE RULE → identify notification

AMENDMENT → identify changed provision and date

GUIDELINE OR SOP → implementation instruction

PORTAL PROCEDURE → digital compliance workflow

CURRENT CLAIM → verify all later changes

SYSTEM / LEVEL / DEFINITION

• BASE RULE → identify notification

• AMENDMENT → identify changed provision and date

STRUCTURE-FUNCTION MECHANISM

• GUIDELINE OR SOP → implementation instruction

• PORTAL PROCEDURE → digital compliance workflow

SCALE / EVIDENCE LIMIT

• CURRENT CLAIM → verify all later changes

• BASE RULE → identify notification

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PORTAL PROCEDURE → digital compliance workflow; CURRENT CLAIM → verify all later changes.

02

STAGE 02

MUNICIPAL RESPONSIBILITY MAP

MUNICIPAL RESPONSIBILITY MAP

SEGREGATE AND HAND OVER

BULK GENERATOR

ADDITIONAL APPLICABLE DUTIES

COLLECTION AND PROCESSING SYSTEM

RUN AUTHORISED FACILITY

SPCB OR PCC

AUTHORISATION AND MONITORING

GENERATOR → segregate and hand over

BULK GENERATOR → additional applicable duties

ULB → collection and processing system

OPERATOR → run authorised facility

SYSTEM / LEVEL / DEFINITION

• GENERATOR → segregate and hand over

• BULK GENERATOR → additional applicable duties

STRUCTURE-FUNCTION MECHANISM

• ULB → collection and processing system

• OPERATOR → run authorised facility

SCALE / EVIDENCE LIMIT

• SPCB OR PCC → authorisation and monitoring

• GENERATOR → segregate and hand over

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OPERATOR → run authorised facility; SPCB OR PCC → authorisation and monitoring.

03

STAGE 03

SEGREGATION-TO-DISPOSAL CHAIN

SEGREGATION-TO-DISPOSAL CHAIN

PREVENT OR REDUCE

AVOID WASTE

PRESERVE RECOVERABLE STREAMS

COLLECT AND TRANSPORT

MOVE WITHOUT REMIXING

RECOVER OR TREAT

PROCESS BY STREAM

PREVENT OR REDUCE → avoid waste

SEGREGATE → preserve recoverable streams

COLLECT AND TRANSPORT → move without remixing

RECOVER OR TREAT → process by stream

RESIDUE → safe final disposal

SYSTEM INPUT / STARTING CONDITION

• PREVENT OR REDUCE → avoid waste

ECOLOGICAL PROCESS / TRANSFER

• COLLECT AND TRANSPORT → move without remixing

OUTCOME / FEEDBACK / LIMIT

• RESIDUE → safe final disposal

- SPCB OR PCC → authorisation and monitoring
- GENERATOR → segregate and hand over

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OPERATOR → run authorised facility; SPCB OR PCC → authorisation and monitoring.

SEGREGATION-TO-DISPOSAL CHAIN PREVENT OR REDUCE AVOID WASTE PRESERVE RECOVERABLE STREAMS COLLECT AND TRANSPORT MOVE WITHOUT REMIXING RECOVER OR TREAT PROCESS BY STREAM

OUTCOME / FEEDBACK / LIMIT

- RESIDUE → safe final disposal
- PREVENT OR REDUCE → avoid waste

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RECOVER OR TREAT → process by stream; RESIDUE → safe final disposal.

PLASTIC ACTOR MATRIX	COVERED PRODUCT OR PACKAGING RESPONSIBILITY	COVERED IMPORTED PRODUCT RESPONSIBILITY	BRAND OWNER	BRANDED PACKAGING RESPONSIBILITY	AUTHORISED END-OF-LIFE ACTIVITY	MUNICIPAL SERVICE ROLE, NOT AUTOMATIC EPR SUBSTITUTE
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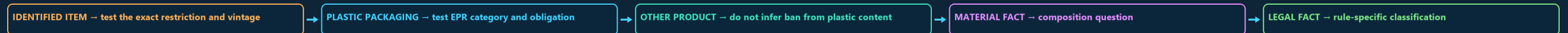
ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
PRODUCER → covered product or packaging responsibility	BRAND OWNER → branded packaging responsibility	ULB → municipal service role, not automatic EPR substitute
IMPORTER → covered imported product responsibility	PROCESSOR → authorised end-of-life activity	PRODUCER → covered product or packaging responsibility

SCALE / STATUS / EXCEPTION

- ULB → municipal service role, not automatic EPR substitute
- PRODUCER → covered product or packaging responsibility

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ULB → municipal service role, not automatic EPR substitute.

BAN-VERSUS-EPR GATE	IDENTIFIED ITEM	TEST THE EXACT RESTRICTION AND VINTAGE	PLASTIC PACKAGING	TEST EPR CATEGORY AND OBLIGATION	OTHER PRODUCT	DO NOT INFER BAN FROM PLASTIC CONTENT	MATERIAL FACT
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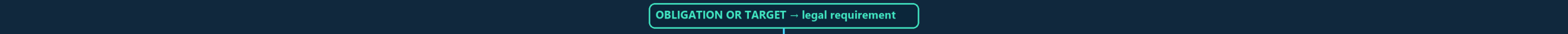


SCALE / EVIDENCE LIMIT

- LEGAL FACT → rule-specific classification
- IDENTIFIED ITEM → test the exact restriction and vintage

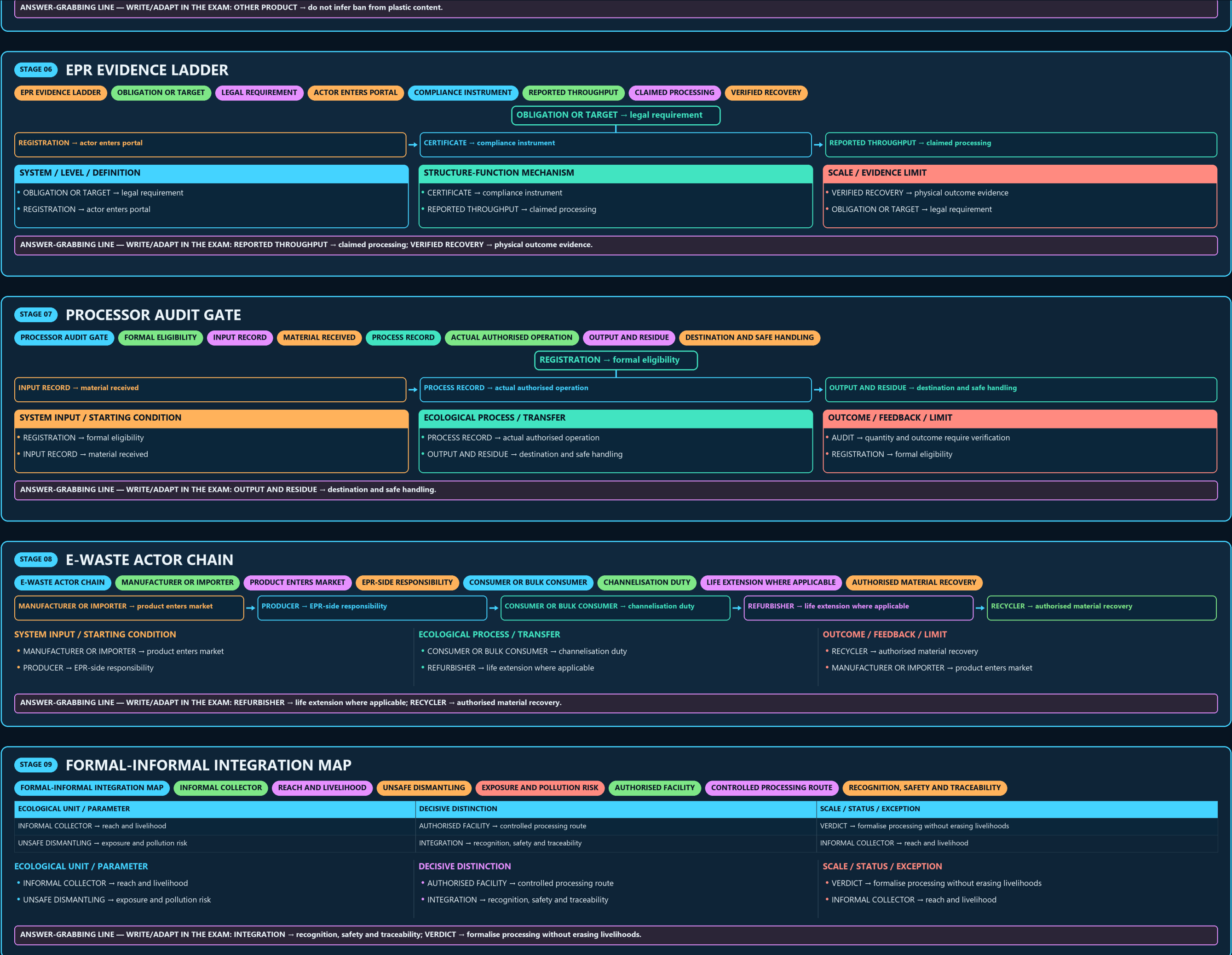
ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: OTHER PRODUCT → do not infer ban from plastic content.

EPR EVIDENCE LADDER OBLIGATION OR TARGET LEGAL REQUIREMENT ACTOR ENTERS PORTAL COMPLIANCE INSTRUMENT REPORTED THROUGHPUT CLAIMED PROCESSING VERIFIED RECOVERY



SCALE / EVIDENCE LIMIT

- VERIFIED RECOVERY → physical outcome evidence
- OBLIGATION OR TARGET → legal requirement



09

STAGE 09

FORMAL-INFORMAL INTEGRATION MAP

FORMAL-INFORMAL INTEGRATION MAP

INFORMAL COLLECTOR

REACH AND LIVELIHOOD

UNSAFE DISMANTLING

EXPOSURE AND POLLUTION RISK

AUTHORISED FACILITY

CONTROLLED PROCESSING ROUTE

RECOGNITION, SAFETY AND TRACEABILITY

ECOLOGICAL UNIT / PARAMETER

INFORMAL COLLECTOR → reach and livelihood

UNSAFE DISMANTLING → exposure and pollution risk

DECISIVE DISTINCTION

AUTHORISED FACILITY → controlled processing route

INTEGRATION → recognition, safety and traceability

SCALE / STATUS / EXCEPTION

VERDICT → formalise processing without erasing livelihoods

INFORMAL COLLECTOR → reach and livelihood

ECOLOGICAL UNIT / PARAMETER

• INFORMAL COLLECTOR → reach and livelihood

• UNSAFE DISMANTLING → exposure and pollution risk

DECISIVE DISTINCTION

• AUTHORISED FACILITY → controlled processing route

• INTEGRATION → recognition, safety and traceability

SCALE / STATUS / EXCEPTION

• VERDICT → formalise processing without erasing livelihoods

• INFORMAL COLLECTOR → reach and livelihood

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: INTEGRATION → recognition, safety and traceability; VERDICT → formalise processing without erasing livelihoods.

10

STAGE 10

STOCK-FLOW AND TECHNOLOGY MATRIX

STOCK-FLOW AND TECHNOLOGY MATRIX

CURRENT WASTE FLOW

EPR AND COLLECTION CONTROLS

LEGACY DUMP OR SITE

REMEDIATION PROBLEM

MATERIAL RECOVERY ROUTE

THERMAL OR CHEMICAL ROUTE

FEEDSTOCK AND EMISSION CONDITIONS

ECOLOGICAL UNIT / PARAMETER

CURRENT WASTE FLOW → EPR and collection controls

LEGACY DUMP OR SITE → remediation problem

DECISIVE DISTINCTION

RECYCLING → material recovery route

THERMAL OR CHEMICAL ROUTE → feedstock and emission conditions

SCALE / STATUS / EXCEPTION

NO UNIVERSAL WINNER → compare complete mass and residue balance

CURRENT WASTE FLOW → EPR and collection controls

ECOLOGICAL UNIT / PARAMETER

• CURRENT WASTE FLOW → EPR and collection controls

• LEGACY DUMP OR SITE → remediation problem

DECISIVE DISTINCTION

• RECYCLING → material recovery route

• THERMAL OR CHEMICAL ROUTE → feedstock and emission conditions

SCALE / STATUS / EXCEPTION

• NO UNIVERSAL WINNER → compare complete mass and residue balance

• CURRENT WASTE FLOW → EPR and collection controls

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NO UNIVERSAL WINNER → compare complete mass and residue balance.

11

STAGE 11

WASTE-GOVERNANCE ANSWER SPINE

WASTE-GOVERNANCE ANSWER SPINE

STREAM, MATERIAL AND RULE VINTAGE

GENERATOR, ULB, PRODUCER, IMPORTER AND PROCESSOR

SEGREGATION, COLLECTION, PROCESSING AND RESIDUE

OBLIGATION, CERTIFICATE, THROUGHPUT AND RECOVERY

INFORMAL SECTOR, LEGACY STOCK, TECHNOLOGY AND PYQ LIMITS

CLASSIFY → stream, material and rule vintage

ASSIGN → generator, ULB, producer, importer and processor

TRACE → segregation, collection, processing and residue

VERIFY → obligation, certificate, throughput and recovery

QUALIFY → informal sector, legacy stock, technology and PYQ limits

SYSTEM / LEVEL / DEFINITION

• CLASSIFY → stream, material and rule vintage

• ASSIGN → generator, ULB, producer, importer and processor

STRUCTURE-FUNCTION MECHANISM

• TRACE → segregation, collection, processing and residue

• VERIFY → obligation, certificate, throughput and recovery

SCALE / EVIDENCE LIMIT

• QUALIFY → informal sector, legacy stock, technology and PYQ limits

• CLASSIFY → stream, material and rule vintage

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: QUALIFY → informal sector, legacy stock, technology and PYQ limits.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

INDIA'S INFORMAL WASTE ECONOMY (WASTE-PICKERS, KABADIWALAS/SCRAP DEALERS, INFORMAL

POLICY TENSION

FORMAL EPR SYSTEMS RISK BYPASSING OR DISPLACING THIS INFORMAL

THIS IS A RECURRING, GENUINELY TWO-SIDED POLICY DILEMMA

PURELY INFORMAL PROCESSING

INDIA'S SINGLE-USE-PLASTIC RESTRICTIONS TARGET SPECIFICALLY IDENTIFIED ITEMS (BASED ON

ENFORCEMENT REALITY

OPTIONAL ECOLOGICAL PROCESS DEPTH

• CAUTION: India's informal waste economy (waste-pickers, kabadiwalas/scrap dealers, informal

• CAUTION: Policy tension: formal EPR systems risk bypassing or displacing this informal

• CAUTION: This is a recurring, genuinely two-sided policy dilemma: purely informal processing

• India's single-use-plastic restrictions target specifically identified items (based on

SCALE, PARAMETER OR STATUS LIMIT

• CAUTION: Enforcement reality: effective enforcement of item-specific bans requires

• CAUTION: National aggregate figures for plastic-waste generation, collection rate and recycling

• CAUTION: E-waste generation growth estimates are typically modelled from electronics-sales data

• CPCB's EPR portals allow producers to meet obligations by acquiring EPR certificates

COMPARATIVE RESTORATION NUANCE

• India's informal waste sector has historically handled a substantial share of actual

• India's single-use-plastic restrictions target specifically identified items rather

• National plastic/e-waste generation and recycling-rate figures are largely modelled/

• NOT: EPR compliance requires a producer's own waste to be physically recycled by a specific

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.